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Neural networks estimation of the density in molecular clouds : Benchmark on Orion B dense cores

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ABSTRACT

Context. /
Aims. This work explores the use of different neural network architectures to estimate the mass-weighted average volume density along the line of sight in molecular clouds, as well as the average density of dense cores. We aim to provide a full benchmark of these architectures, highlighting their advantages and limitations in this physical context.
Methods. We use synthetic datasets generated from magnetohydrodynamic simulations from which column and average volume density maps were derived. These datasets allows the training of different neural networks: a CNN (UNet) and two conditioned generative ones: cINNs and DDPMs. We apply differents numericals and physicals metrics to test the robustness, confidence and limitations of each method.
Results. /
Conclusions. /

Key words. ISM: clouds – ISM: structures – stars: formation – machine learning – neural networks – mass function

1. Introduction

2. Methods

In this work, because our goal is to study dense structures in molecular clouds, such as prestellar cores, we choose to estimate, using neural networks, the mass-weighted average volume density along the line of sight:

$$\langle n_{\rm H}\rangle_{\rm m}(x,y) = \frac{\int_0^L n_H(x,y,z)m(x,y,z)dz}{\int_0^L m(x,y,z)dz}, \quad (1)$$

from the column density map derived from dust emission:

$$N_{\rm H}(x,y) = \int_0^L n_H(x,y,z)dz, \quad (2)$$

where x and y are the plane-of-sky positions, z is the position in line of sight axis, L the cloud length, $n_{\rm H}$ the hydrogen volume density, $m(x,y,z) = n_{\rm H}(x,y,z)\mu m_{\rm H}$.

Thus, our benchmark focuses on a 2D → 2D learning problem. Note also that the mass-weighted average density can be converted into the average density of dense cores using a small correction (see Appendix A).

Except for making the simulations, all steps, from creating the datasets and training neural networks to applying them to observations, are performed using our code `POLARIScore`[1] ,which includes our implementations of network architectures using `PyTorch` (Paszke et al. 2019).

2.1. MHD Simulations and Datasets

Neural networks need to be trained on synthetic data, where they can learn the relevant correlations and underlying physics. For segmentation tasks, such training data can consist of human-generated or algorithm-generated labels on observations (see, for example, filament-identification studies (Alina et al. 2022)).
However, in our task of estimating a physical quantity in molecular clouds, we do not have access to true observational values. This is where simulations become essential, since they provide full access to all required physical quantities.

The properties of the simulations used to train the neural network are therefore extremely important and directly influence the network's performance. **talk about properties of the cloud Orion B**

These datasets are created using magnetohydrodynamical (MHD) simulations of a star-forming molecular cloud: *ORION* (Ntormousi & Hennebelle 2019), publicly available through the GALACTICA[2] database and shown in Figure 1.
These simulations are isothermals, model MHD turbulence and have a size of 66.1 pc with an initial resolution of 0.01 pc, reaching a few hundred AU using an AMR grid. The simulations were performed using RAMSES (Teyssier 2002), with self-gravity applied, star-formation treated using Lagrangian sink particles and ideal MHD resolved and no chemistry.

To facilitate data manipulation, we use a 1024^3 density cube with a width of 25.8pc, centered at the simulation's midpoint, with a resolution of 0.026pc per pixel. To increase the amount of training data, projections along the three axes are used as shown in Figure 1.
Then, images with a resolution of 128×128 pixels and a physical

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[1] `https://github.com/ZackRibeiro/POLARIScore`
[2] `http://www.galactica-simulations.eu/db/STAR_FORM/ORION/`

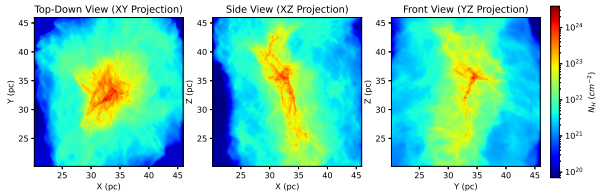

Fig. 1: Column densities for 3 axes of the same molecular cloud from the low magnetic field ORION simulation. (Ntormousi & Hennebelle 2019)

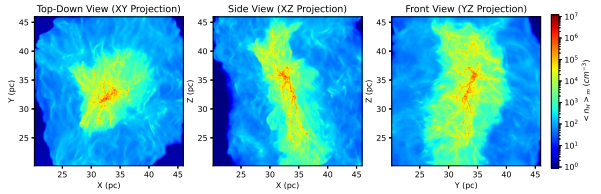

Fig. 2: Mass-weighted average volume densities for 3 axes of the same molecular cloud from the low magnetic field ORION simulation. (Ntormousi & Hennebelle 2019)

size of 3.33pc are randomly cropped from the three faces. An image is retained in the dataset if it meets several conditions:

- It is spatially distant enough from an already explored region, meaning it does not overlap with an area covered by an existing image.
- It does not contain a disproportionately large fraction of very low-density regions, i.e, the edges of the simulation (WNM).
- It exhibits significant density variations within the explored regions. A score is based on the smoothness of the image using Laplacian of the density, favoring regions with filaments over those with nearly homogeneous density. (See Appendix D)

The generated dataset is then randomly shuffled and split between training set (70%-80%) and validation set (30%-20%). For example, using only the simulation shown in Figure 1, we obtain a total of $\approx$ 100 covered regions, each represented by a pair of images: (column density, volume density)

2.2. Training

Training a neural network is a crucial step in achieving good performance. There are multiple choices that we need to make to obtain optimal results.

- We want the model, with N free parameters (weights), to converge to a global minimum of the loss function rather than getting stuck in a local well. Therefore, the choice of optimizer, which performs backward error propagation (adjusting the model's weights), is important. Here, we use ADAM (Kingma & Ba 2017).
- Loss function: A well-chosen loss function is essential for accurately representing the problem. Here, we generally use the mean squared error or minimizing the negative log-likelihood depending on the architecture.
- Learning rate: The learning rate is another critical factor, it determines the step size the optimizer takes in the loss function during each update (epoch). Ideally, it should decrease when the loss reaches a plateau. To manage this, we use a *scheduler*.

Each epoch, i.e., learning step, the model processes all the dataset, column density images, and predicts their mass-weighted density. A way to artificially increase the size of the training batch is to apply random transformations to the images at each epoch (Yang et al. 2023), such as random rotations and random vertical or horizontal flips (no deformations). Afterward, the prediction loss is computed using the chosen loss method, and backward error propagation is performed. Then, a new epoch begins, repeating the same steps.

2.3. Machine learning

In this benchmark, we explore three different neural network architectures. We start with a robust and widely used model for image-to-image regression tasks, the UNet (Ronneberger et al. 2015; Oktay et al. 2018). We then test more recent state-of-the-art conditioned generative approaches, namely Conditional Invertible Neural Networks (cINNs) (Ardizzone et al. 2021) and Diffusion Models (Ho et al. 2020).

We do not include Generative Adversarial Networks (GANs) (Goodfellow et al. 2014) in this work mainly because they are known to suffer from mode collapse (Kossale et al. 2022).

2.3.1. Convolutional Neural Network

Convolutional Neural Networks (CNNs) were introduced (Fukushima (1980), LeCun et al. (1989)) as a specialized architecture for grid-like data such as images or maps. Instead of connectiong every input to every neuron (as in multi-layer perceptrons), CNNs restrict connections to local receptive fields. Each neuron in a convolutional layer is connected only to a small region of the input, and the same set of learnable weights (a filter or kernel) is shared across different spatial locations. This enables CNNs to learn spatially invariant features such as edges, textures and structure morphologies.

Mathematically, for a 2D input X (e.g a column density region), the convolution operation with a filter W is given by:

$$Y_{i,j} = f(\sum_{m=1}^{M}\sum_{n=1}^{N} W_{m,n} \cdot X_{i+m,j+n} + b), \quad (3)$$

where W is the learnable convolutional kernel, X is the input feature map, Y is the output feature map, f is a nonlinear activation function such as ReLU, b is the learnable bias term.

In this work, we are using an Attention-U-Net (Ronneberger et al. (2015),Oktay et al. (2018)) architecture, shown in Figure 3. It features a symmetric encoder-decoder structure where the encoder captures high-level features of the column density map through convolution (Eq 3) and max pooling operations, while the decoder gradually reconstructs the mass-weighted density map using upsampling layers.
A key feature of U-Net is its *skip connections*, which allow direct information flow between corresponding encoder and decoder layers, preserving spatial details lost during downsampling. **small sentence on Attention blocks ?**

2.3.2. Conditional Invertible Neural Network

However, state-of-art networks used today are not deterministic architectures such as UNet, but stochastic models that sample from probability distributions.

The goal of these networks is to sample from the conditional data distribution $p_{\text{data}}(X|c)$ where X is the target random vari-

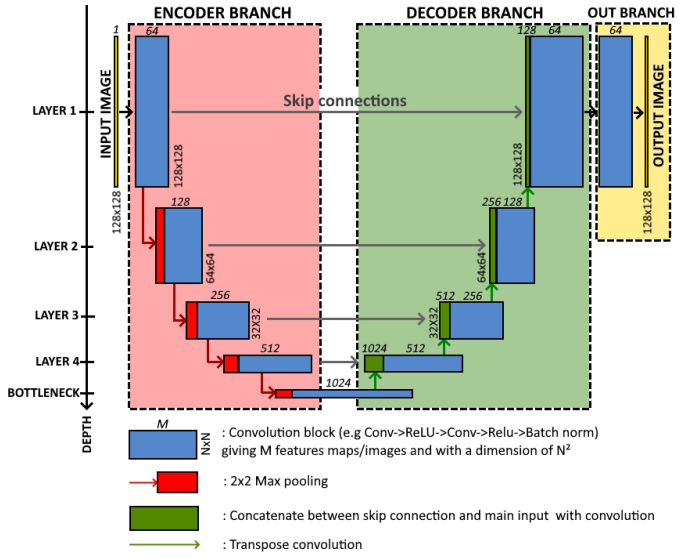

Fig. 3: U-Net architecture (Ronneberger et al. 2015). Each blue-box corresponds to a convolutionnal block. The number of channels is denoted on top of the box. The x-y-size is provided at the right or left edge of the box.

able (here $\langle n_{\rm H} \rangle_{\rm m}$) and c the conditioning information (from observable, here $N_{\rm H}$). Ff the inverse problem is ill-posed, i.e if one cannot uniquely infer the mass-weighted average volume density map $\langle n_{\rm H} \rangle_{\rm m}(x, y)$ from the 2D column-density map $N_{\rm H}(x, y)$, the generative network instead learns to sample physically plausible solutions drawn from the full posterior distribution

$$p(\langle n_{\rm H} \rangle_{\rm m}(x, y) | N_{\rm H}(x, y)). \tag{4}$$

To make sampling tractable, we map the unknown complex distribution to a simple latent one, typically a standard multivariate Gaussian:

$$p_z(Z) = N(0, I), \tag{5}$$

where Z represents the latent variables.

The network then learns a bijective transformation f between X and Z, conditioned on c:

$$Z = f(X, c), \; X = f^{-1}(Z, c) \tag{6}$$

The approach used in conditional invertible neural networks (cINN, Ardizzone et al. (2021)) is to train *the forward direction*

$$f : X \to Z \tag{7}$$

such that the pushforward of the data distribution becomes Gaussian:

$$f_*(p_{\rm data}(X|c)) = p_z(Z). \tag{8}$$

This is achieved through normalizing flows (Rezende & Mohamed 2016), which transform a complex distribution into a simple one via a sequence of invertible transformations: $f_1, ..., f_K$:

$$Z = f_K \cdot f_{K-1} \cdot ... \cdot f_1(X, c). \tag{9}$$

For each invertible transformation f_k, the change-of-variable formula gives the exact likelihood:

$$p_{\rm data}(X|c) = p_z(f(X, c)) |\det \frac{\partial f(X, c)}{\partial X}|. \tag{10}$$

During training, the cINN maximizes the log-likelihood of the data under this model. At inference time, we draw $Z \sim N(0, I)$ and obtain samples of the physical solution via the inverse mapping: $X = f^{-1}(Z|c)$.

To ensure reasonable computation time, each transformation f_k in the flow must be easily invertible, and its Jacobian determinant must be efficient to compute. cINNs achieve this by using conditional coupling blocks (CCBs). Let the input of a coupling block be split (in features channels) as $u \rightleftarrows (u_1, u_2)$, and the output be $v \rightleftarrows (v_1, v_2)$. A typical forward transformation of a CCB is:

$$\begin{cases} v_1 = u_1 \odot \exp(s_1(u_2, c)) + t_1(u_2, c), \\ v_2 = u_2 \odot \exp(s_2(v_1, c)) + t_2(v_1, c), \end{cases} \tag{11}$$

where s_i, t_i are scale and translation functions implemented by sub-networks (e.g MLPs). These sub-networks do not need to be invertible because they are only evaluated in the forward pass; the coupling block itself remains analytically invertible. The inverse transformation is obtained by algebraic rearrangement of Equation 11:

$$\begin{cases} u_2 = (v_2 - t_2(v_1)) \odot \exp(-s_2(v_1)) \\ u_1 = (v_1 - t_1(u_2)) \odot \exp(-s_1(u_2)) \end{cases} \tag{12}$$

Because each coupling block is triangular in structure, its Jacobian is also triangular. Thus the log-determinant of the Jacobian is simply the sum of the scaling terms:

$$\log |\det J| = \sum s_1(u_2, c) + \sum s_2(v_1, c), \tag{13}$$

where the sums run over features and spatial dimensions. This makes CCBs computationally efficient while retaining exact invertibility and tractable likelihood.

Some words on architecture with figure (Encoder, Haar wavelet downsampling)

2.3.3. Denoising Diffusion Probabilistic Model

Denoising Diffusion Probabilistic Models (DDPMs, Ho et al. (2020)) generate samples by gradually transforming white noise, i.e a sample from the latent Gaussian prior $p_z = N(0, I)$, into a data sample $x_0 \sim p_{\rm data}$. This is achieved through a sequence of learned denoising steps parameterized by a neural network, typically a UNet.

The forward diffusion process progressively destroys structure by adding Gaussian noise with a predefined variance schedule β_t. In the continous-time limit, this process is described by the stochastic differential equation (Itô SDE):

$$dx = f(x, t)dt + g(t)dW_t = -\frac{1}{2}\beta(t)x_t dt + \sqrt{\beta(t)}dW_t, \tag{14}$$

where W_t is a Wiener process. Here:

- $g(t) = \sqrt{\beta(t)}$ is the diffusion strength,
- $f(x, t) = -\frac{1}{2}\beta(t)x$ is the drift term,
- $t \in [0, T]$ parametrized the diffusion trajectory.

This SDE is a variant of the Ornstein-Uhlenbeck process, whose solutions converge to a Gaussian distribution (Uhlenbeck & Ornstein 1930). The process is variance-preserving, sice as $t \to \infty$, $x_t \sim N(0, I)$.

To generate data, DDPMs simulate the reverse-time SDE:

$$dx = [f(x,t) - g(t)^2 \nabla_{x_t} \log p_t(x_t)]dt + g(t)d\overline{W}_t, \tag{15}$$

where $\overline{W}_t$ is a Wiener process running backward in time. The quantity

$$\nabla_{x_t} \log p_t(x_t) \tag{16}$$

is the score function, i.e the gradient of the log-density of the noisy data at time t. Because the exact score is intractable, the network learns an approximation $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$.

In DDPM parameterization, the neural network instead predicts the noise

$$s_\theta(x_t, t) \approx \frac{\epsilon_\theta(x_t, t)}{\sigma_t}, \tag{17}$$

where $\sigma_t = \sqrt{1 - \overline{\alpha_t}}$ is the standard deviation of the forward noising kernel.

The forward noising process has the closed-form solution:

$$x_t = \sqrt{\bar{\alpha}_t}, x_0 + \sqrt{1 - \bar{\alpha}_t}, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{18}$$

where $\overline{\alpha_t} = \prod_{i=1}^{t}(1 - \beta_i)$ encodes the cumulative noise schedule.

Thus, training a DDPM consists of learning to predict the injected noise ϵ, which in turn provides the score needed to solve the reverse SDE and sample realistic data from pure Gaussian noise.

In the conditional setting, the observable $N_{\rm H}$ is incorporated by concatenating it with the noisy sample x_t at the input of the UNet, ensuring that each denoising step is conditioned on the available physical information.

Some words on architecture with figure (Encoder, Noise scheduling)

Salimans & Ho (2022)

2.4. Application

The trained neural networks are applied to Orion B, a giant molecular cloud located at a distance of approximately 400 pc (Menten et al. 2007). Herschel observations (Könyves et al. 2020; André et al. 2010) reveal a fragmented filamentary structure, with star formation primarily concentrated in dense clumps.

The networks are applied to the column density map derived from dust emission. Prestellar cores are identified using multiwavelength source extraction and spectral energy distribution (SED) fitting of the dust emission, and are taken from the catalogue of Könyves et al. (2020).

Because the neural networks are trained on regions with a physical size of 3.3 pc, the column density map is downsampled by a factor of 4.4 to match the spatial resolution of the training dataset. The observation is then divided into overlapping patches of 128×128 pixels, each of which is provided independently as input to the neural network.

The patches are allowed to overlap in order to provide multiple contextual environments for each position in the map and to reduce edge effects inherent to convolutional architectures. The final prediction at each pixel is obtained by averaging the outputs from all overlapping patches covering that pixel.

In addition of the neural networks, we also apply a parametric fit of the form

$$f(x) = \sum_{i=1}^{4} a_i \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right), \tag{19}$$

derived from simulations, to model the conditional distribution $\langle n_{\rm H} \rangle_{\rm m}\,(N_{\rm H})$ (see Fig. 4).

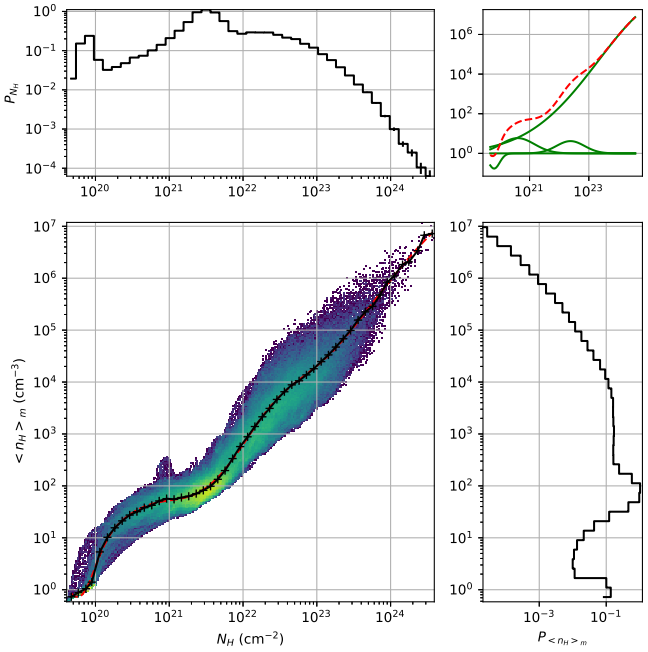

Fig. 4: Gaussian-mixture fit of the simulation correlation between $\langle n_{\rm H} \rangle_{\rm m}$ and $N_{\rm H}$ (bottom left). The top and right panels show the cumulative probability distributions of $\langle n_{\rm H} \rangle_{\rm m}$ and $N_{\rm H}$, respectively. The top-right panel displays the individual Gaussian components (green) and their sum (red).

3. Results

3.1. Benchmark on simulation data

In this section, we focus on the metrics evaluated using predictions on the validation set, which is a subset of the training data generated with the ORION simulation Ntormousi & Hennebelle (2019). Two representative predicted regions are shown in Fig. 5.

During the training of all neural networks, we use the mean squared error (MSE) computed in logarithmic space to monitor the generalization performance of the models. Specifically, the MSE loss is evaluated on the validation set and is defined as

$$\mathrm{MSE} = \mathbb{E}\left[\left(\log Y_{\rm pred} - \log Y_{\rm sim}\right)^2\right], \tag{20}$$

where $Y_{\rm pred}$ and $Y_{\rm sim}$ denote the predicted and simulated (true) mass-weighted average volume-density maps, respectively.

If one considers only the MSE values reported in Table 1, UNet (0.19) and DDPM (0.25) appear to be the most performant models, ahead of the cINN (0.50). However, the MSE is an average over all regions and environments and therefore does not fully capture the ability of a network to adapt to different physical conditions. In particular, because of its averaging nature, a

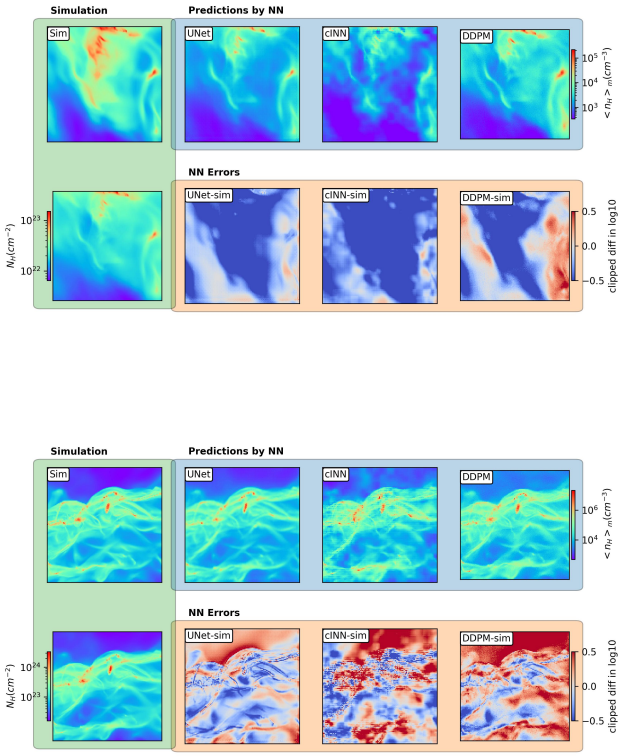

Fig. 5: Neural network validation set predictions on two regions of the validation dataset (3.33pc of width)

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model may achieve a low MSE by performing well in diffuse background regions while still exhibiting significant errors in filamentary structures, whose spatial filling factor is much smaller than that of the cold neutral medium (CNM).

We also measure the accuracy which is the fraction of pixels with an absolute error below a given threshold σ:

$$\mathrm{Acc}(\sigma) = \langle \mathrm{Acc}_i \rangle_{\mathrm{batch}}(\sigma) = \frac{1}{B}\sum_{i}^{B} \frac{N\left(|Y_{\mathrm{pred},i} - Y_{\mathrm{sim},i}| < \sigma\right)}{N_{\mathrm{pixels}}}, \quad (21)$$

B is the number of regions in the validation set, and $N(\cdot)$ denotes the number of pixels in the maps Y that satisfy the specified condition.

To diagnose whether the network underestimates high densities or low densities, and on which density interval the network works well, we plot the residuals curves. Note that the means of the residuals bins are moved to 0 by using post-prediction correction when we apply the model to observations (see Appendix B).

Network accuracies (Fig. 6) and residuals (Fig. 7) reveal that all networks systematically underestimate certain density ranges. In particular, UNet underestimates high densities, while cINN and DDPM predominantly underestimate intermediate densities around $n_{\mathrm{H}} \sim 10^4\,\mathrm{cm}^{-3}$, which are typical of dense cores. As a consequence, when applied to observational data without corrections, these models are expected to underestimate dense-core densities.

We also find that the prediction errors increase with density. This trend can be explained by the fact that the networks primarily learn the diffuse background at relatively low densities, as

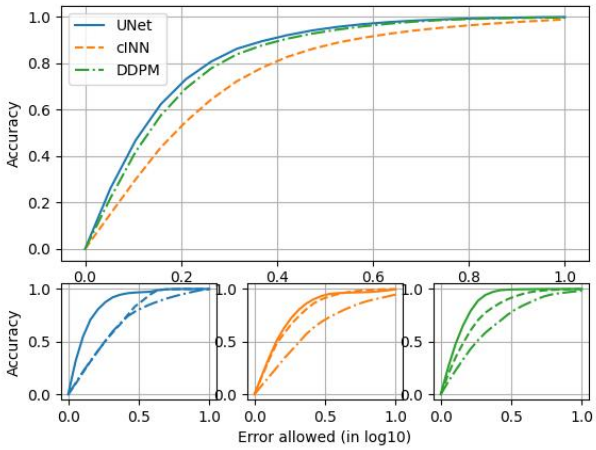


Fig. 6: Accuracy of the neural networks: UNet (blue), cINN (orange), and DDPM (green). Top panel: overall accuracy comparison between the networks. Bottom panels: accuracy curves for each network, binned by true mass-weighted average volume density (solid line: $n_{\mathrm{H}} \leq 200$ cm^{-3}; dashed line: 200 cm^{-3} $< n_{\mathrm{H}} \leq 10^4$ cm^{-3}; dash-dotted line: $n_{\mathrm{H}} > 10^4$ cm^{-3}).

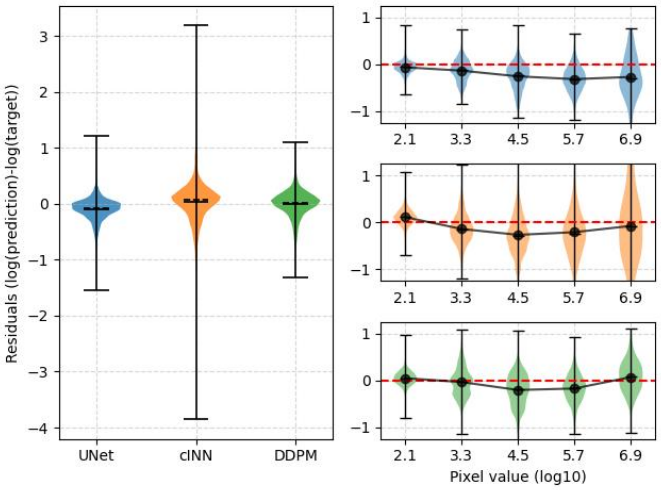


Fig. 7: Logspace residuals of the neural networks: UNet (blue), cINN (orange), and DDPM (green). Left panel: overall logspace residuals on all the density ranges, colored vertical shape shows the residual distributions and black lines are the distribution extremums. Right panel: Logspace residuals for each network, binned by true mass-weighted average volume density.

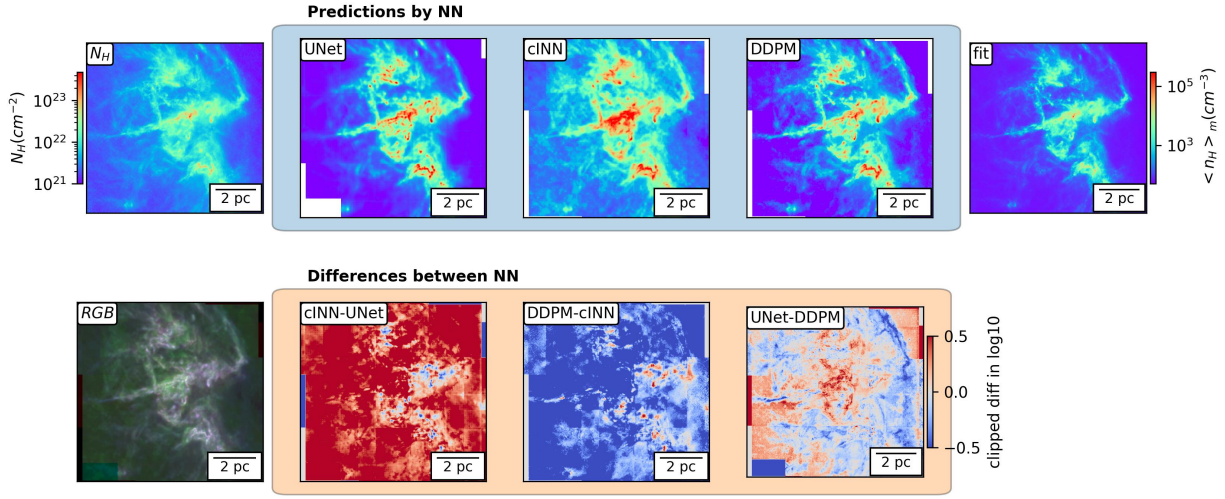
dense filaments and cores occupy a much smaller fraction of the total volume and are therefore underrepresented in the training data even with our dataset preparation.

This behaviour is also visible in the spatial error maps shown in Fig. 5, which further highlight network-specific artifacts. For UNet (and partially masked in the other models by larger-scale artifacts), small-scale convolutional artifacts are visible. The cINN exhibits characteristic stride patterns, which may indicate early signs of non-invertibility and suggest that some information is missing to accurately infer the mass-weighted average volume density from the column-density map alone. Finally, the

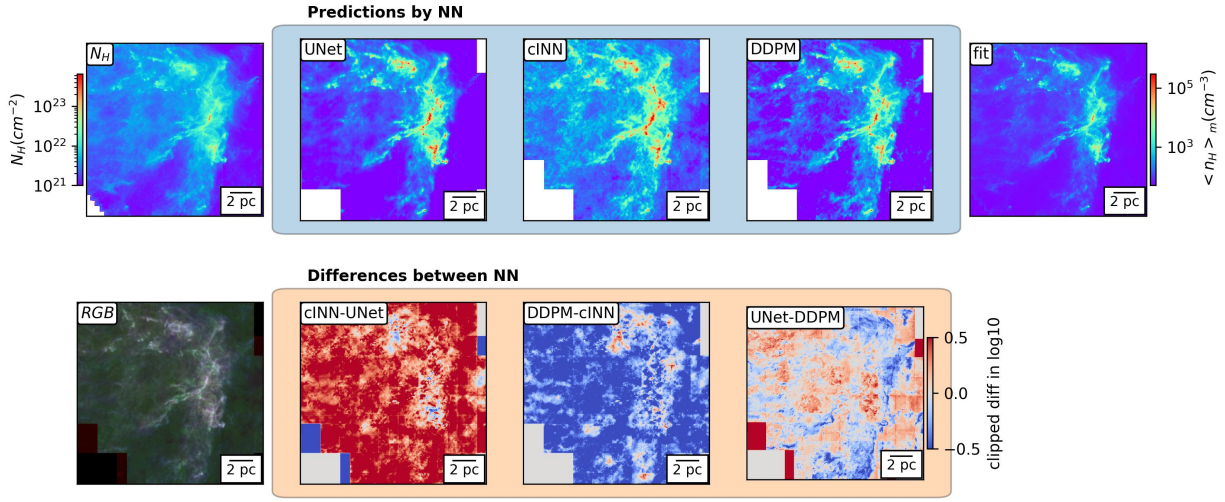
DDPM displays a white-noise like artifact pattern, which is inherent to the stochastic nature of the method.

Metric	Model		
	UNet	cINN	DDPM
MSE (log10)	0.19	0.50	0.25
Parameters ($\times 10^7$)	13	3.8	8.2
Inference speed (reg/s)	3.5	3.4	0.32
Error at 80% accuracy	...	...	...
Training stability	High	Low	Mid
Required training-set size	Low	...	...

Table 1: Neural-network model metrics.



(a) NGC2068 and NGC2071 regions.



(b) Flame Nebula and The Cloak regions.

Fig. 8: Neural network predictions on Orion B. Top row (left to right): the column-density map derived from dust emission, followed by the predictions of the three neural networks (UNet, cINN, and DDPM), and finally the estimate obtained from the Gaussian fit (Eq. 19). Bottom row: an RGB composite image constructed from the three neural-network predictions (in the same order as above), followed by clipped logarithmic difference maps between the neural-network predictions.

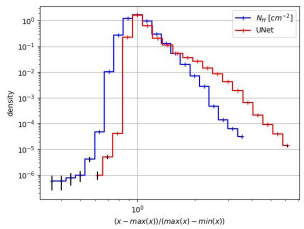
3.2. Tests on Orion B

The purpose of this section is not to derive new physical results for the Orion B molecular cloud, but rather to assess the consistency of the model predictions with existing observational studies. Predictions for Orion B are shown in Fig. 8, where we focus on two major star-forming subregions. The difference maps between the model predictions, also shown in Fig. 8, highlight that the cINN systematically predicts higher densities in more diffuse regions.

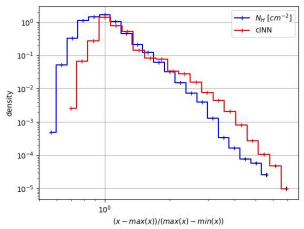
3.2.1. Large-scale structures

The density probability distribution functions (PDFs) of molecular clouds are known to follow a lognormal shape at low densities, with a power-law tail at high densities, both for column

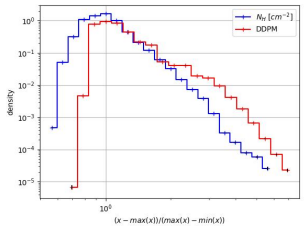
density and volume density (**?**). However, for the mass-weighted average volume-density PDF, we observe an excess at high densities, appearing as a bump in the power-law regime at $n_{\rm H} \sim \ldots$.



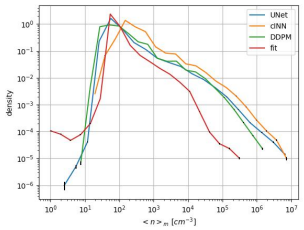
(a) $N_{\rm H}$ pdf with UNet $\langle n_{\rm H} \rangle_{\rm m}$ pdf



(b) $N_{\rm H}$ pdf with cINN $\langle n_{\rm H} \rangle_{\rm m}$ pdf



(c) $N_{\rm H}$ pdf with DDPM $\langle n_{\rm H} \rangle_{\rm m}$ pdf



(d) All networks $\langle n_{\rm H} \rangle_{\rm m}$ distributions

Fig. 9: Density distributions. First three panels: comparison between the column-density PDF and the $n_{\rm H}$ PDF, with the horizontal axis normalized. Last panel: distributions of all model predictions.

3.2.2. Prestellar cores

- **Density PDFs:** Comparison of the probability distribution functions of the predicted densities (but how ?).
- **Turb ?**
- **Fractal dimension:** Using the relation $P = A^{D/2}$, where P is the perimeter and A the area, we test whether the structures follow a power-law scaling, implying a constant fractal dimension D across spatial scales.
- **Typical size of structures (?):** Evaluate characteristic spatial scales (e.g., whether filament widths cluster around $\sim$ 0.1 pc).
- **Comparison with the core catalog of Könyves et al.:** Differences between predicted core densities and catalogued values.
- **Core morphology:** Visual inspection to determine whether predicted cores are embedded within density clumps or appear isolated (hierarchical picture).
- **Robustness to resolution changes:** Evaluate how the predictions vary when the observational resolution is modified, using both the validation set and real observations.
- $N_{\rm H}(n_{\rm H})$ **relation (dense cores only):** Fit the predicted column-volume density of dense cores relation to infer plausible temperatures or sound speeds.
- **Dense Core Mass Function (DCMF):** Agreement with Könyves et al. regarding peak position, width, low-mass cutoff, and high-mass power-law slope.

4. Discussions

Other methods to find mass-weighted density : Gaches & Grudić (2024), Orkisz & Kainulainen (2024).

5. Conclusions

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Appendix A: From mass-weighted averages to core densities

In this work, the neural networks aim to predict the mass-weighted average volume density:

$$\langle n_{\rm H}\rangle_{\rm m}(x,y) = \frac{\int_0^L n_H(x,y,z)m(x,y,z)dz}{\int_0^L m(x,y,z)dz}, \tag{A.1}$$

where x and y are the plane-of-sky positions, z is the position in line of sight axis, L the cloud length, $n_{\rm H}$ the hydrogen volume density, $m(x,y,z) = n_{\rm H}(x,y,z)\mu m_H$.

If we discretize the density along the l.o.s into M components with constant volume density in it (e.g for a datacube simulation, M corresponds to the number of cells), then Eq A.1 becomes

$$\langle n_{\rm H}\rangle_{\rm m} = \frac{\sum_{i=1}^{M} n_{{\rm H},i}^2 \times L_i \mu m_{\rm H}}{\sum_{i=0}^{M} n_{{\rm H},i} \times L_i \mu m_{\rm H}} = \frac{\sum_{i=1}^{M} n_{{\rm H},i}^2 L_i}{\sum_{i=0}^{M} n_{{\rm H},i} L_i}, \tag{A.2}$$

where L_i is the length of component i. This quantity is chosen mainly because this weighted average density is close to the average density of dense structures along the l.o.s such as dense cores.

However, when a dense core is embedded by a very diffuse region (i.e outside filmentary structures), a correction is required to convert the mass-weighted average density $\langle n_{\rm H}\rangle_{\rm m}$ into the true average volume density of the dense core $n_{{\rm H},c}$. To simplify notation, we denote the average core density simply by n_c and volume densities without the $_{\rm H}$.

Consider two components along the line of sight ($M = 2$):

- A diffuse medium with length L_d and with constant density n_d.
- The dense core with length $L_c = 2r_c$ where r_c is the core radius, and with constant density n_c.

Equation A.2 then reduces to

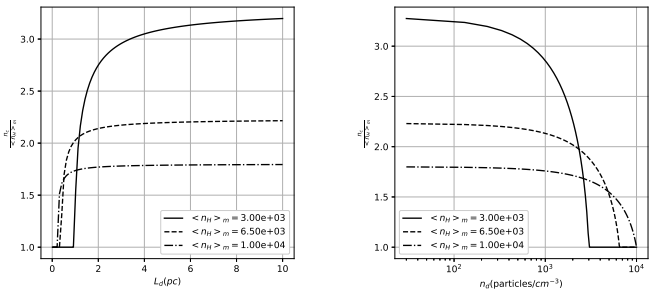
$$\langle n_{\rm H}\rangle_{\rm m} = \frac{n_c^2 L_c + n_d^2 L_d}{n_c L_c + n_d L_d}, \tag{A.3}$$

To estimate n_c, two parameters must be assumed: n_d and L_d, while L_c is derived from observations as two times the core radius. The column density: $n_{\rm H} = n_c L_c + n_d L_d$ removes one degree of freedom, giving

$$n_c = N_{\rm H}\left(\frac{1+\sqrt{1-(\frac{L_d}{L_c}+1)(1-\frac{\langle n_{\rm H}\rangle_{\rm m} L_d}{N_{\rm H}})}}{(L_c+L_d)}\right), \text{ if } L_d \text{ assumed.} \tag{A.4}$$

$$\text{or } n_c = \frac{n_d}{2}\left(1+\sqrt{1-\frac{4N_{\rm H}}{L_c n_d}(1-\frac{\langle n_{\rm H}\rangle_{\rm m}}{n_d})}\right), \text{ if } n_d \text{ assumed.} \tag{A.5}$$

But these corrections can lead to $\langle n_{\rm H}\rangle_{\rm m}$ being higher than n_c, because the conditions $n_d > 0$ or $L_d > 0$ are not enforced. To rectify this, we clamp the core density as $n_c = \max(n_c, \langle n_{\rm H}\rangle_{\rm m})$, with n_c computed from one of the previous formulas.



(a) L_d free (Eq. A.4). (b) n_d free (Eq. A.5).

Fig. A.1: Correction $\langle n_{\rm H}\rangle_{\rm m} \to n_c$ for different $\langle n_{\rm H}\rangle_{\rm m}$; the vertical axis shows the ratio $n_c/\langle n_{\rm H}\rangle_{\rm m}$. The used parameters are $r_c = 0.05$ pc and $N_{\rm H} = 10^{22}$ cm^{-2}.

If one wants to assume specific values for L_d and n_d:

$$n_c = \frac{\langle n_{\rm H}\rangle_{\rm m}}{2}\left(1+\sqrt{1+\frac{4L_d n_d}{L_c\langle n_{\rm H}\rangle_{\rm m}}(1-\frac{n_d}{\langle n_{\rm H}\rangle_{\rm m}})}\right) \tag{A.6}$$

Figure A.1 displays the functions $n_c(L_d)$ (left) and $n_c(n_d)$ (right) for various values of $\langle n_{\rm H}\rangle_{\rm m}$. A notable feature is that n_c rapidly converges toward an upper limit when L_d becomes large or when n_d is small. The limits are

$$\lim_{L_d\to+\infty} n_c = \lim_{n_d\to 0} n_c = \sqrt{\frac{\langle n_{\rm H}\rangle_{\rm m} N_{\rm H}}{L_c}}, \tag{A.7}$$

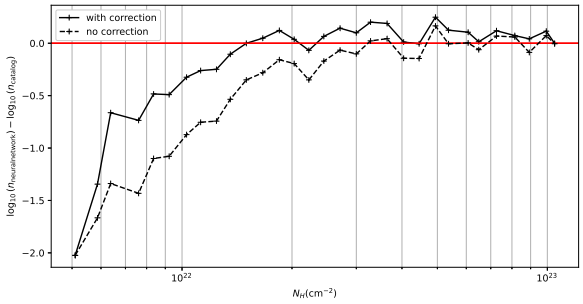


Fig. A.2: Differences in log space between the catalog core densities and the neural-network-predicted densities as a function of column density. The cloud is Orion B, neural network is UNet, and the dense-core catalog is from Könyves et al. (2020). The thick black line shows the density difference using the correction (Eq. A.4: $L_d = 10$pc); the dotted black line shows the result without the correction. The red horizontal line indicates zero difference, i.e., where the neural-network prediction matches the catalog density.

We see in Figure A.2 that the correction indeed increases the dense-core density by an order of magnitude in diffuse regions, i.e., at low column densities. However, at high column densities, the two-medium approximation no longer holds, and therefore we can define an upper threshold, for example 3×10^{22}cm^{-2}, above which the density correction should not be applied.

Appendix B: Validation error and bias

Residuals distributions; Correction of the bias; Residuals error propagation to metrics (MC);

Appendix C: U-Net architecture variations

C.1. U-Net with Kolmogorov Arnold Networks

C.2. Size and Context embedding in U-Net

Appendix D: Score-based training set

Creating a training dataset is a crucial step in building an effective model. In the particular case of molecular clouds structures, we want to avoid overtraining on the borders of the simulation (WNM), as these regions contain less valuable information for our objective compared to the dense regions with filaments and cores. To achieve this, we implement an additional filtering step:

- We compute a "*score*" s(image) for each new candidate area beffore adding to the dataset. This score is calculated using a custom function that aligns with our training objectives.
- We then apply a filter function $f(s)$. If a randomly generated number r in the range $[0, 1]$ satisfies $r < f(s)$, the region is accepted and added to the training set. I.e $f(s)$ is the probability to accept and add an area with a score of s in the dataset.

This approach ensures that the dataset focuses more on important structures (filaments and cores) rather than sparse or less informative regions.

We can also use multiple scoring functions, each capturing different aspects of the dataset, and combine them using weighted (w_i) sums to account for various criteria. Mathematically, the final score s can be expressed as:

$$s = \sum_i s_i w_I \tag{D.1}$$

This score can be made for the column density image $s(N_{\rm H})$ and the volume density image $s(\langle n_{\rm H}\rangle_{\rm m})$.

D.1. Score functions

Smoothness score The score function used when generating our training set needs to be based on the smoothness of the selected area. This is because if an area is entirely within a diffuse region, it will exhibit low spatial frequency variation (i.e, only large-scale structures) and lack filaments.
To ensure that each selected image contains both diffuse regions and some filaments, we use the variance of the smoothness as a criterion. The smoothness itself is defined as the Laplacian of the image, which highlights areas with sharp density changes (such as filaments).

$$s_\Delta(x) = \mathrm{Var}[\Delta(\log(1+x) - \min[\log(1+x)])], \tag{D.2}$$

where x is the pixel value. The variance in equation D.2 is to get a mix of dense and high-density regions.

Difference score A basic criterion that can be chosen is the difference between the normalized column density and the normalized average volume density (Eq D.3). This helps identify regions where the local density distribution differs significantly between the two representations.

For example, when using mass-weighted average density, this criterion will highlight dense regions, since areas with high volume density but relatively lower column density indicate localized dense structures along the l.o.s.

$$s_{\rm ref} = \mathrm{Var}[\frac{N_{\rm H} - \min(N_{\rm H})}{\max(N_{\rm H}) - \min(N_{\rm H})} - \frac{\langle n_{\rm H}\rangle_{\rm m} - \min(\langle n_{\rm H}\rangle_{\rm m})}{\max(\langle n_{\rm H}\rangle_{\rm m}) - \min(\langle n_{\rm H}\rangle_{\rm m})}] \tag{D.3}$$

D.2. Filter functions

As a filter function $f(s)$, we use a sigmoid:

$$f(s) = \frac{1}{1 + \exp(-A(s - B))}, \tag{D.4}$$

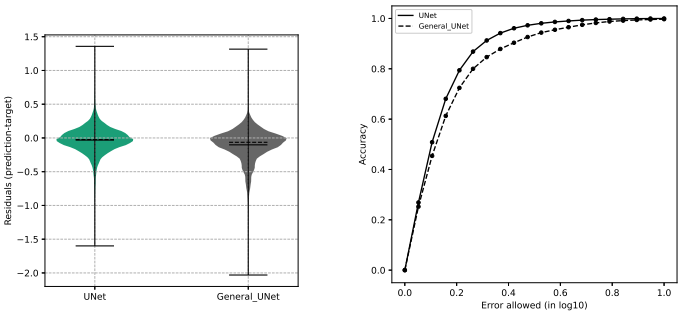
where A is the step length and B the offset, and are chosen empirically.

D.3. Network robustness to training-set variations

Here, we test the effect of excluding WNM or large homogeneous regions from the training set for some of our architectures. We compare the model performances on two different datasets:

- One dataset is the same as used in this paper, containing approximately 100 regions of the ORION simulation, excluding WNM and favoring heterogeneous regions in the selection.
- The other dataset contains 170 regions, including all WNM regions from the simulation, with no selection bias toward heterogeneity.

U-Net test



(a) Residuals (b) Accuracy

Fig. D.1

cINN **train a cINN with less data**

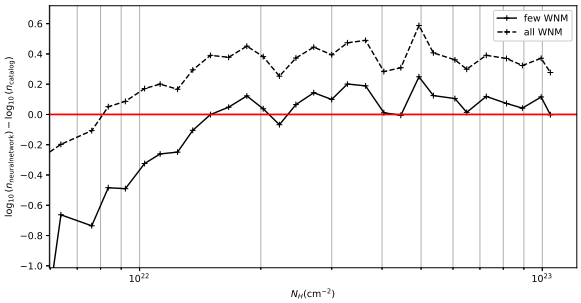


Fig. D.2: Differences in log space between the catalog core densities and the neural-network-predicted densities as a function of column density. Cloud is Orion B, model is UNet, and the dense-core catalog is from Könyves et al. (2020). Thick black line: dataset without WNM; dotted black line: dataset with WNM. The red horizontal line indicates zero difference.

Appendix E: Examples of dense core predictions

Figure: horizontal: Powerlaw, Unet, cINN, DDPM vertical: top: column density, bot: average volume density.
One or few for differents environments: low column density, high column density, in filaments, outside filaments...